

Case Analysis in Isabelle/HOL

```
theory CaseAnalysis
  imports Main
begin
```

Another standard way to prove facts is by *case analysis*. In this method you prove a theorem by showing that the theorem's statement holds for a finite number of cases. Those cases need to be exhaustive, i.e. at least one of has to hold.

In the following example, the proof is by considering two cases: $x \geq y$ and $\neg x \geq y$. This is done in the step case.

```
lemma lemma4: " $a \leq b \implies \text{Suc } a \leq \text{Suc } b$ "
  apply(subst Suc_le_mono)
  .
```

```
lemma "( $x::\text{nat}$ ) - y \leq x"
proof(induction x)
  case 0
  have " $0 - y = 0$ "
  using zero_diff
  .
  also have " $\dots \leq 0$ "
  using order_refl
  .
  finally show ?case
  .
next
case (Suc x)
show ?case
proof(cases " $x \geq y$ ")
  case True
  have 1: " $(\text{Suc } x) - y = \text{Suc } (x - y)$ "
  using Suc_diff_le[OF True]
  .
  have " $\text{Suc } x - y \leq \text{Suc } x - y$ "
  using order_refl
  .
  then have 3: " $\text{Suc } x - y \leq \text{Suc } (x - y)$ "
  apply(subst 1[symmetric])
  .
  also have " $\dots \leq \text{Suc } x$ "
  using lemma4[OF Suc.IH]
  .
  finally show ?thesis
  .
next
case False
have 1: " $x < y$ "
```

```

      using not_le_imp_less[OF False]
    .
  have 2: " $Suc\ x \leq y$ "
    using Suc_leI[OF 1]
  .
  have 3: " $Suc\ x - y = 0$ "
    using diff_is_0_eq[OF 2]
  .
  have 4: " $0 \leq Suc\ x$ "
    using zero_le
  .
  show ?thesis
    using 4
    apply(subst 3)
  .
qed
qed

end

```